

ANDREW ZHANG

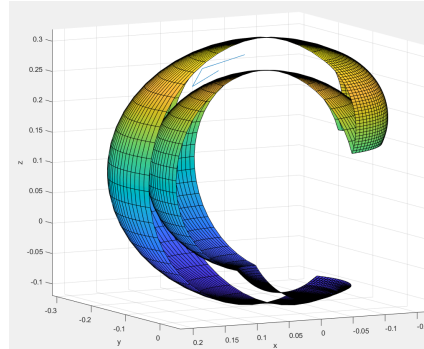
MSc Robotics Engineering, National University of Singapore
BASc Mechanical Engineering with Mechatronics Option, University of Waterloo

NAO Robot Arm Trajectory Generation and Motion Control

University of Waterloo - Robotics Dynamics and Control Course



NAO v6 robot



Limits of right hand end effector in 3D Cartesian space, and four points defining a tested trajectory



Desired vs real trajectories of robot arm in x, y, z

What:

- Controlled the end-effector of a NAO V6 robotic arm to follow a complex trajectory accurately, with 3 DOF (Degrees Of Freedom).

How:

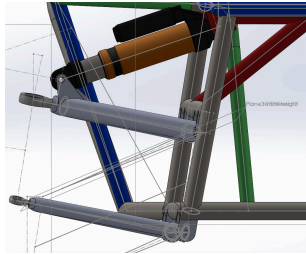
- End effector limits were identified with forward kinematics, and appropriate target points were selected. A minimum jerk trajectory was generated.
- First-pass joint torques were determined with an inverse dynamics control and an arm simulation. The gains were refined and the joint torques were optimized with the real robotic arm.

Result:

- Accuracy requirements were met, and the arm successfully followed the trajectory.

Front Suspension Design for Off-Road Vehicle

University of Waterloo - Baja SAE Team



Suspension, front view



Support structure

What:

- Designed front suspension of an all terrain vehicle to withstand an instantaneous force of 3200 N per wheel with a yield safety factor of 1.2.
- Numerous constraints - building on pre-existing structure, accommodating future parts, competition rules and more.

How:

- Extremes of the possible suspension positions were identified to derive resultant design load cases.
- Used Solidworks FEA to simulate preliminary designs under design loading conditions.
- This led to the addition of a support structure to prevent yielding and refinements to facilitate ease of manufacturing.

Result:

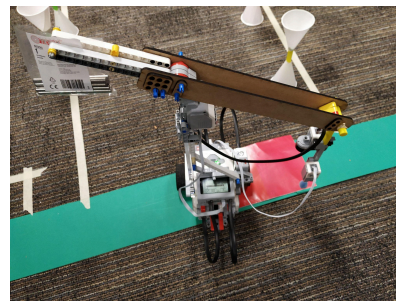
- Final design withstood loads while also satisfying all constraints.

Flower Pollinator Robot Prototype

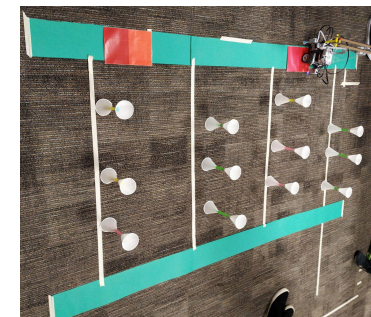
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Pollinator robot, side/bottom



Pollinator robot, above/behind



Pollinator robot on a trial course flower bed, with fake flowers of varying heights

What:

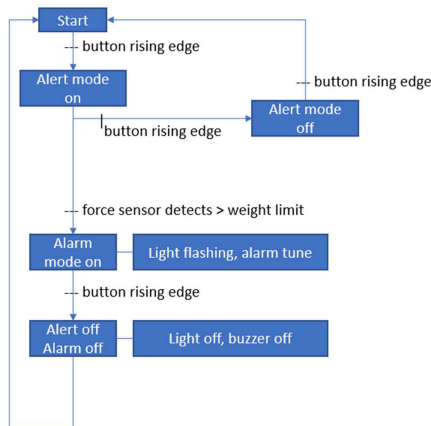
- Created a LEGO® MINDSTORMS® EV3 robot to pollinate flowers.

How:

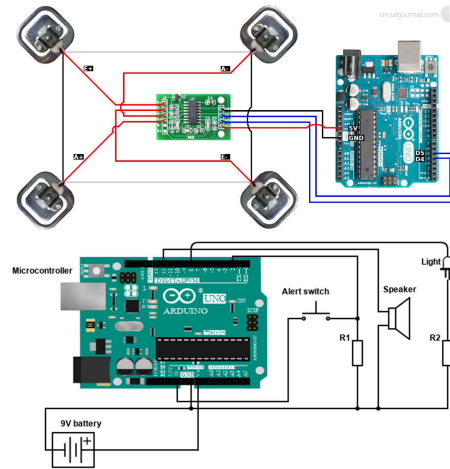
- The robot moved forward with two motors until an ultrasonic sensor detected a flower.
- The robot arm lowered the pollen attachment until a touch sensor contacted the flower. A color sensor was used to detect the bounds of the flower bed.

Kelp Harvester Alert System Prototype

University of Waterloo - Capstone Project



Simplified state diagram



Load cell setup (top), circuit exclusive of load cells (bottom)



Harvester prototype

What:

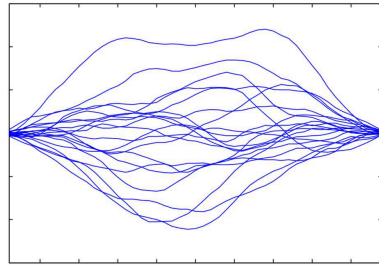
- Created a kelp harvesting system designed for installation on boats, with an alert system for kelp bin fullness.

How:

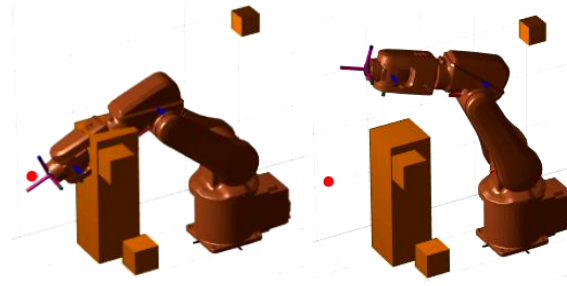
- An Arduino UNO microcontroller was programmed in C++ to process sensor data and provide control signals.
- Four load cells detected bin weight. They were wired, then soldered to a voltage signal amplifier. Pins were soldered to the amplifier to enable breadboard prototyping.
- LEDs and buzzers connected to the system indicated “ready” and “threshold weight detected” states.

Implementing STOMP Algorithm for Robot Arm Obstacle Avoidance

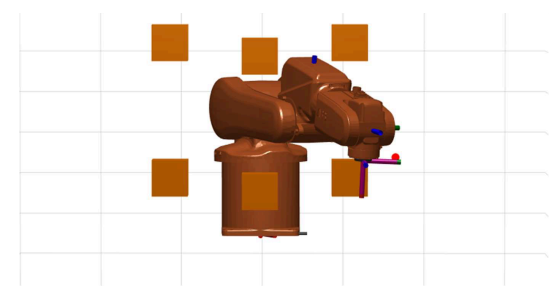
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Variance of a joint path in STOMP algorithm



Arm on partially optimized trajectory colliding with object (left), fully optimized trajectory (right)



Robot arm completing second obstacle scenario

What:

- Moved the end effector of an ABB IRB 6 DOF MATLAB robot model between positions while avoiding collisions with obstacles.

How:

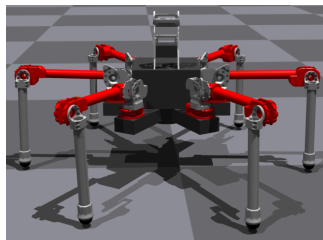
- Achieved by implementing STOMP (Stochastic Trajectory Optimization for Motion Planning) algorithm to find optimized trajectories.

Result:

- The arm successfully completed two obstacle avoidance scenarios, which included additional constraints on end effector position and orientation.

Training Hexapod Robot with Machine Learning

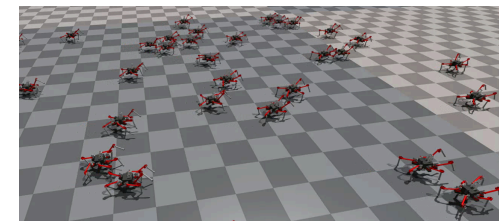
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Hexapod robot standing in Isaac Gym physics simulation



Hexapod robot ready to push an object



Multiple hexapod robots training in parallel

What:

- Trained a six-legged robot to walk on four legs, with two legs positioned to push an object in an Isaac Gym physics simulation.

How:

- This was accomplished via reinforcement learning, programmed with Python and run on a Linux OS.
- After testing, four layer structures were used for the actor and critic neural networks.

Result:

- The robot was able to walk to the goal positions with its two pushing limbs posed correctly.